

Analytic Geometry

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math).

1. **[L1 · Slope]** The coefficient of x is the slope: $m = 6$.
2. **[L1 · Graphing]** Slope = 1; y-intercept (0, -5).
3. **[L1 · Parallel]** Parallel lines share the slope: $m = 5$.
4. **[L1 · Perpendicular]** Negative reciprocal of 4 is $-1/4$.
5. **[L1 · Length]** $d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
6. **[L1 · Midpoint]** $M = ((2+8)/2, (2+6)/2) = (5, 4)$.
7. **[L2 · Slope]** $m = (7 - (-1))/(6 - 2) = 8/4 = 2$.
8. **[L2 · Graphing]** y-intercept (0, 8); x-intercept: $0 = -2x + 8 \rightarrow x = 4 \rightarrow (4, 0)$.
9. **[L2 · Parallel]** Same slope 4, y-intercept 3: $y = 4x + 3$.
10. **[L2 · Perpendicular]** Negative reciprocal of $2/3$ is $-3/2$.
11. **[L2 · Length]** $d = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$.
12. **[L2 · Midpoint]** $M = ((-3+7)/2, (5-1)/2) = (2, 2)$.
13. **[L3 · Slope]** $(k - 4)/(5 - 1) = 3$, so $k - 4 = 12$, giving $k = 16$.
14. **[L3 · Graphing]** $-2y = -4x + 8$, so $y = 2x - 4$.
Slope = 2; y-intercept (0, -4).
15. **[L3 · Parallel]** Slope = $(10 - 1)/(3 - 0) = 9/3 = 3$, equal to 3, so yes — the lines are parallel.
16. **[L3 · Perpendicular]** Slopes are 3 and $-1/3$; product = $3 \times (-1/3) = -1$, so yes — they are perpendicular.
17. **[L3 · Length]** $AB = 4$ (vertical), $AC = 3$ (horizontal), $BC = \sqrt{3^2 + 4^2} = 5$.
Perimeter = $4 + 3 + 5 = 12$.
18. **[L3 · Midpoint]** $Q = (2 \cdot 3 - (-1), 2 \cdot (-2) - 4) = (7, -8)$.